Final Project Proposal — AI Study Assistant

Team: Akmurzina Mariza, Kim Ilana

1. Problem and Users

University students spend too much time navigating lengthy lecture notes and PDFs when preparing for exams. Our AI Study Assistant helps students quickly find relevant information using Retrieval-Augmented Generation (RAG) and automatically generates practice quizzes.

2. AI Features and Evaluation

Students upload PDF lecture notes, ask questions, and generate quizzes. We will evaluate the RAG system using 30 test questions with reference answers, measuring answer correctness and source relevance. Our target is at least 80% accuracy.

3. Agentic Coding Tools

We plan to use Claude Code (paid) and OpenAI Codex (free and paid access) for planning, code generation, debugging, and testing. We will document agent sessions and important decisions.

4. MVP and Stretch Goals

MVP: PDF upload, RAG-based question answering with source references, quiz generation, and a simple Streamlit interface.

Stretch goals: Automatic lecture summaries and personalized study recommendations.

5. Data, APIs, and Access

Python, Streamlit, ChromaDB, and an LLM API. Data will come from our own university lecture PDFs and sample educational materials. Streamlit and ChromaDB are freely available; LLM API access will be configured using available free or paid credits.

6. Team Responsibilities

Akmurzina Mariza: PDF processing, text embeddings, ChromaDB integration, document retrieval, and RAG evaluation.

Kim Ilana: Streamlit interface, LLM API integration, AI-generated answers, quiz generation, and testing.

Both members: Agentic coding tools, GitHub commits, documentation, debugging, and final presentation.